ZOMBIE SMASHER 3D

GAME MECHANICS AND ENVIRONMENT DESIGN

PROJECT OVERVIEW

GAME CONCEPT

A 3D zombie endless runner played through a third-person camera behind the survivor. The player runs down three lanes, dodging obstacles and zombies while collecting coins and power-ups.

PROJECT DIRECTION

Four environment identities: Forest, Desert, Spooky Dark City and Abandoned City. Each changes the setting and atmosphere while the core run-and-survive gameplay stays the same.

DEVELOPMENT PLATFORM

Built in Unity 6.6 (6000.6.1f1) with the Universal Render Pipeline and C#. A custom Editor setup wizard creates the scenes, materials, prefabs and build settings.

PROJECT TEAM MEMBER IDs

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CORE GAMEPLAY LOOP

1. RUN
The survivor runs forward automatically on a three-lane track that streams in 40 m chunks.

2. DODGE
Switch lanes, jump or slide to avoid obstacles and the zombies in the way.

3. COLLECT
Grab coins and power-ups to raise the score and gain short-lived advantages.

4. SURVIVE
Stay ahead of the pursuing horde as speed and zombie density keep rising.

3	11 → 25 m/s	+ bonus	3
lanes	speed ramp with distance	every 500 m	playable survivors

Score = distance + coins, with the best score saved separately for each theme. Survivors: Ranger, Medic and Mechanic.

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PLAYER CONTROLS

ACTION	KEYBOARD	GAMEPAD	TOUCH / MOUSE
Move left	A or Left arrow	D-pad left	Swipe left
Move right	D or Right arrow	D-pad right	Swipe right
Jump	W, Up arrow or Space	South button (A / Cross)	Swipe up
Slide / fast-fall	S or Down arrow	East button (B / Circle)	Swipe down
Pause	Esc or P	Start	HUD pause button

INPUT BUFFERING

Jump and slide presses are buffered for 0.15 s, so a slightly early press still counts.

INPUT SYSTEMS

Works with both the new Input System and the legacy Input Manager, on PC and Android.

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THE FOUR GAME THEMES

DENSE ZOMBIE FOREST

Muddy trail through thick woods, drifting fog and light shafts breaking through the canopy.

POST-APOCALYPTIC DESERT

Cracked desert road, sandstorms, heat-hazed sky and wrecked military convoys.

SPOOKY DARK CITY

Night streets with rain, flickering neon signs and reflective wet asphalt.

ABANDONED APOCALYPTIC CITY

Hazy ruined buildings, burning fires and columns of smoke rising over the street.

Every theme has its own 6 obstacle types, scenery, lighting, sky, fog, post-processing, music, ambience and footstep sounds.

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OBSTACLES AND DIFFICULTY

LOW OBSTACLES

Jump over them. Hitting one makes the survivor stumble.

HIGH OBSTACLES

Slide under them. Hitting one makes the survivor stumble.

FULL-HEIGHT BLOCKS

Change lanes around them. Hitting a block ends the run.

STUMBLE AND THE HORDE

- A stumble brings the pursuing horde right behind you for 6 seconds.
- A second stumble in that window lets the horde catch you.
- Running into a zombie is also fatal.

DIFFICULTY SCALING

- Run speed rises from 11 to 25 m/s.
- Zombie density, faster zombie types and obstacle complexity grow with distance.
- A distance bonus is awarded every 500 m.

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ZOMBIE TYPES

WALKERS
Slow, shambling zombies that block a lane.

FAST RUNNERS
Quick zombies that close in and lunge.

BRUTES
Large, heavy zombies with a loud roar.

HORDES
Groups of zombies spread across lanes.

EMERGERS
Rise from the ground or step out of doorways.

PURSUING HORDE
A horde follows the player throughout the run. Stumbling pulls it in close, so every mistake raises the pressure.

ZOMBIE AI
Zombies move along the lanes with steering, lunges and attack ranges, which suits a procedurally generated straight track.

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POWER-UPS

MAGNET	SPEED BOOST	SHIELD	DOUBLE COINS	HEALTH PROTECTION
10 s	**5 s**	**8 s**	**12 s**	**Until used**
Pulls nearby coins toward the player.	Runs faster and is invulnerable.	Smashes straight through anything.	Every coin counts twice.	Absorbs one fatal hit.

Active power-ups appear on the HUD with timers, and each one has its own pickup sound and colour.

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USER INTERFACE AND SCENES

MainMenu → ThemeSelection → Forest / Desert / DarkCity / AbandonedCity Level

MENUS

- Play and Quick Run
- Theme selection cards with best score
- Character select (3 survivors)
- Settings: sound, quality, reset progress

IN-GAME HUD

- Score, distance and multiplier
- Coin counter
- Active power-up timers
- Progress bar to next safe zone
- Pause button

PAUSE AND GAME OVER

- Resume, restart or main menu
- Final score, distance, coins and best score
- Run again or change theme

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SETUP WIZARD AND TECH STACK

TOOLS > ZOMBIE RUNNER > SETUP WIZARD

- Checks URP and the input backend
- Creates folders, URP materials and prefabs
- Builds the 6 scenes and adds them to Build Settings
- Applies player settings (Android IL2CPP, ARM64, landscape)
- Validates the project and prints a colour-coded log
- Safe to run again after importing new assets

TECH STACK

- Unity 6.6 (6000.6.1f1), Universal Render Pipeline
- C# scripts (about 8,600 lines), uGUI, Input System
- Procedural meshes, textures and synthesised audio
- Object pooling and chunk streaming
- Floating origin every 2000 m
- Low / Medium / High / Auto quality tiers
- Builds for Android (APK) and Windows

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CONCLUSION

WHAT WE BUILT

- Third-person three-lane zombie endless runner
- Four distinct themed worlds
- Five zombie types and five power-ups
- Complete menu, HUD and scene flow

FUTURE SCOPE

- Import free CC0 models for a more realistic look
- Play-test and tune difficulty balance
- Profile frame rate on mobile devices
- Add more themes and survivors

THANK YOU

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